H2 Geography Paper 2 Suggested Answers

Cluster 4 Fieldwork

1 Five Singaporean students investigated the success of reusable bags initiative in supermarkets as a response to climate change on a Sunday afternoon. They came up with the following hypothesis:

'Supermarket shoppers in Singapore are likely to bring their own reusable bags when charged a fee for disposable carrier bags'.

To test their hypothesis, the students collected data at two supermarket outlets located within the same neighbourhood. Supermarket Outlet A imposes an additional charge of $0.05 for each disposable carrier bag, while Supermarket Outlet B does not. The students split themselves into two groups and collected data concurrently at both outlets. At each outlet, the students stood near the checkout counters to record the total number of shoppers who purchased groceries, and the number of shoppers who brought their own reusable bags. They also took photographs of some shoppers at the checkout area.

With the help of their family members, they also recruited 5 residents living in the neighbourhood to complete an online survey.

Resource 1 is a map showing the locations of Supermarket Outlets A and B. Resources 2A and 2B show photographs taken by students at Supermarket Outlets A and B. Resource 3 shows the online survey form used by the students. Resource 4 shows the data collected at each supermarket outlet.

(a) Suggest why the hypothesis is appropriate for the investigation. [4]

Award 1 mark for each suggestion on how the hypothesis is appropriate, to a maximum of 4 marks. Award a maximum of 1 additional mark for further development of each suggestion, where applicable.

Explanation [1m] + Further Development [+1m]
• Clearly defined variables make the hypothesis appropriate because it identifies the independent variable as whether shoppers are *charged a fee for disposable carrier bags* and the dependent variable as whether they *bring their own reusable bags*.
• Measurability and testability make the hypothesis appropriate because students can count the total number of grocery shoppers and the number bringing reusable bags.

(b) Using Resource 1, explain the advantages of choosing the two supermarket outlets for data collection. [4]

Award 1 mark for each explanation on the advantages of choosing the two supermarket outlets for data collection, to a maximum of 4 marks. Award a maximum of 1 additional mark for further development of each explanation, where applicable.

Explanation [1m] + Further Development [+1m]
• Accessibility is an advantage and both sites convenient to reach during fieldwork.
• Transport connections can bring a steady flow of shoppers/greater footfall to both outlets and enable the students to collect a sufficiently large sample size.

(c) With reference to Resources 2A, 2B and 3 explain possible ethical issues arising from the students' data collection process. [6]

Award 1 mark for each explanation on the possible ethical issues arising from the students' data collection process to a maximum of 6 marks. Award a maximum of 1 additional mark for further development of each explanation, where applicable.

Explanation [1m] + Further Development [+1m]
• Lack of informed consent for photographs is a possible ethical issue because the resources show *clearly identifiable shoppers and supermarket employees*, yet there is no evidence that permission was obtained before photographing them.
• Absence of informed consent is a possible ethical issue because the survey does not explain the investigation's purpose, how responses will be used, or the respondent's right to withdraw.
• Lack of anonymity and confidentiality is a possible ethical issue because the survey requires respondents to provide their *name, mobile phone number, age and address*, allowing individual survey responses to be traced directly to them.

(d) With reference to Resource 3, explain two strengths and one limitation of the online survey form. [6]

Award 1 mark for each explanation on the strengths and limitation of the online survey form to a maximum of 6 marks. Award a maximum of 1 additional mark for further development of each explanation, where applicable. Two strengths and one limitation is required for 6 marks.

Explanation [1m] + Further Development [+1m]
• Standardised response categories are a strength because the survey form provides every respondent with identical options, with little space for subjectivity.
• A combination of quantitative and qualitative data is a strength because it gathers measurable responses in Questions (i) to (v), while Question (vi) allows respondents to explain their reasoning.
• Restrictive response options are a limitation because Question (v) ilists only four strategies and provides neither an *"Others"* option nor an option indicating that none is effective.

(e) Explain how the students could improve their sampling method for the online survey. [6]

Award 1 mark for each explanation on how the students could improve their sampling method for the online survey, to a maximum of 6 marks. Award a maximum of 1 additional mark for further development of each explanation, where applicable.

Explanation [1m] + Further Development [+1m]
• A larger sample size would improve the online survey because responses from only five residents is inadequate.
• An improved sampling method (i.e. probability sampling) should replace recruitment through family members.
• Sampling from more neighbourhoods would improve the testability of the hypothesis because residents from only one neighbourhood have been investigated.

(f) Describe how the data shown in Resource 4 may be represented. [4]

Award 1 mark for each description on how the data may be represented, to a maximum of 4 marks.
Award a maximum of 1 additional mark for further development of each description, where applicable.

Description [1m]
• A compound bar graph may be used, using one stacked bar for each supermarket outlet, with each bar divided into shoppers bringing disposable carrier bags and reusable bags.
• The horizontal axis should show *Supermarket Outlets A and B*, while the vertical axis should show the number of shoppers.
• The data should be plotted accurately, with Outlet A divided into *16 disposable-bag shoppers and 36 reusable-bag shoppers*, and Outlet B divided into *19 shoppers for each category*.
• A descriptive title, clearly labelled axes, numerical values and a legend distinguishing the two bag categories should be included, enabling the data from both outlets to be interpreted accurately.

(g) With reference to all the Resources, evaluate the validity of the students' fieldwork. [10]

Levels marked using **Generic Level Descriptors for H2 fieldwork evaluative questions**.

Levels	Marks	Generic Level Descriptors for H2 Fieldwork Evaluative Question
3	9–10	Evaluation is analytical and coherent. Response addresses the question and demonstrates good knowledge and understanding of fieldwork methods relevant to the given context.
2	5–8	Evaluation is mostly analytical and coherent. Weaker responses in this level will have evaluation that is broadly analytical and generally coherent. Response generally addresses the question and demonstrates adequate knowledge and understanding of fieldwork methods relevant to the given context.
1	1–4	Response is descriptive with limited or no analysis and evaluation. Response is fragmented and lacks clarity. Response lacks focus on the question and may be largely irrelevant to it. Response shows basic knowledge and understanding of fieldwork methods. Response has some, though limited, relevance to the given context.
0	0	No creditworthy response

Section B

Cluster 3 Sustainable Future and Climate Change

2 Resource 5 shows the global surface temperature changes in 2017. Resource 6 shows the condition of coral reefs in Mexico from 2018 to 2022. Resource 7A shows women involved in road jobs in Dhaka, Bangladesh. Resource 7B shows the women-only train carriage in Tokyo, Japan.

(a) With reference to Resource 5, describe the global surface temperature changes in 2017. [4]

Award 1 mark for each description of the global surface temperature changes in 2017, to a maximum of 4 marks. Award a maximum of 1 additional mark for further development of each description, where applicable.

Description [1m] / Further development [+1m]
• There is predominantly positive temperature change in 2017.
• The highest temperature increases are across the Arctic, northern Canada and northern Russia.
• The lowest temperature changes are in the North Atlantic south of Greenland, where a distinct area cooled by approximately *0.2°C to 0.5°C* instead of experiencing warming.

(b) Explain possible reasons for the global surface temperature changes shown in Resource 5. [5]

Award 1 mark for each explanation on the possible reasons for the global surface temperature changes, to a maximum of 5 marks. Award a maximum of 1 additional mark for further development of each explanation, where applicable.

Explanation [1m] + Further Development [+1m]
• The enhanced greenhouse effect from greenhouse-gas emissions explains the widespread warming because additional carbon dioxide and methane absorb more outgoing longwave radiation, increasing surface temperatures by approximately *0.2°C to 2.0°C* across most regions.
• The reduction of carbon sinks explains some warming in Resource 5 because deforestation reduces carbon dioxide removal through photosynthesis, allowing atmospheric concentrations to rise and strengthening the enhanced greenhouse effect.
• The ice-albedo effect explains the strong Arctic warming because melting snow and ice expose darker surfaces with lower albedo, increasing incoming shortwave radiation absorption and causing further temperature rises.

(c) With reference to Resource 6, describe the changes in the condition of coral reefs in Mexico from 2018 to 2022. [4]

Award 1 mark for each description on the changes in the condition of coral reefs in Mexico, to a maximum of 4 marks. Award a maximum of 1 additional mark for further development of each description, where applicable.

Description [1m] / Further development [+1m]
• There is an overall deterioration followed by partial recovery.
• The greatest negative change involved critical-condition reefs whose proportion rose sharply.
• There is partial improvement from 2020 to 2022, as critical-condition reefs declined, while fair-condition reefs increased.

(d) Explain how changes in aquatic ecosystems as a result of contemporary climate change may impact humans. [6]

Award 1 mark for each explanation on how changes in aquatic ecosystems as a result of contemporary climate change may impact humans, to a maximum of 6 marks. Award a maximum of 1 additional mark for further development of each explanation, where applicable.

Explanation [1m] + Further Development [+1m]
• The impact on supporting ecosystem services occurs when warming, acidification and coral bleaching damage nursery habitats/ reduce primary productivity/ disrupt nutrient cycling, weakening aquatic food webs that sustain human populations.
• The impact on regulating ecosystem services occurs when climate change damages coral reefs, mangroves and seagrass beds, weakening coastal protection against wave energy, storm surges, erosion and coastal flooding.
• The impact on provisioning ecosystem services occurs when warmer and more acidic waters reduce fish reproduction or shift species towards cooler areas, lowering seafood supplies, fishing incomes and household food security.

(e) Explain why gender equality is important for achieving liveable cities. [4]

Award 1 mark for each explanation on why gender equality is important for achieving liveable cities, to a maximum of 4 marks. Award a maximum of 1 additional mark for further development of each explanation, where applicable.

Explanation [1m] + Further Development [+1m]
• Women's equal access to education, employment, childcare and finance increases women's workforce participation, reduces urban poverty and raises household quality of life.
• Safe transport, accessible public spaces and equal healthcare access strengthen women's personal security, social participation, sense of belonging and overall urban well-being.

	(f)	Using Resources 7A and 7B, explain the issues faced by women living in cities.	[6]

Award 1 mark for each explanation on the issues faced by women living in cities, to a maximum of 6 marks. Award a maximum of 1 additional mark for further development of each explanation, where applicable.

Explanation [1m] + Further Development [+1m]
• W*omen involved in road jobs in Dhaka* may depend on physically demanding, low-skilled employment, indicating limited access to secure and better-paid urban occupations.
• The provision of a *women-only train carriage in Tokyo* suggests women may face harassment, intimidation or insecurity when travelling on ordinary public transport.
• Women undertaking roadside work in Dhaka are exposed directly to traffic, dust, noise and outdoor heat, creating significant occupational health and safety risks.

Section C

3 Evaluate the impact of urban reimaging on sustainable urban development. [20]

Possible Approaches:

Candidates could approach the question by making a judgment on whether urban reimaging has a overall positive or negative impact through a balanced consideration of its economic, social, and environmental dimensions. Candidates could evaluate how successful reimaging strategies (e.g., flagship developments, culture-led regeneration, or mega-event hosting) drive economic revitalisation and environmental renewal, while weighing these against negative trade-offs such as social displacement or increased ecological footprints. Candidates could further analyse the interdependence of these dimensions—such as how short-term economic gains from reimaging may come at the expense of long-term social cohesion or environmental durability.

Candidates could also approach the question by making a judgment on whether the impact of urban reimaging varies depending on context, through a consideration of two or more case studies (e.g., post-industrial cities vs. global hubs, or cities in the Global North vs. Global South). Candidates could analyse how contextual factors—such as the quality of urban governance, stakeholder involvement, financial backing, and existing infrastructure—dictate whether reimaging achieves a balanced, sustainable outcome or exacerbates spatial and socio-economic inequalities.

Levels marked using Generic Level Descriptors for H2 essays

4 'Firms play a key role in climate change response.' Evaluate this statement. [20]

Possible Approaches:

Candidates could approach the question by making a judgment on whether firms play the *primary* or most crucial role in climate change response compared to other key actors, namely states and non-governmental organisations (NGOs). Candidates could evaluate the unique capacity of firms to drive innovation, invest capital into green technologies, and transform global supply chains. This could be weighed against the essential role of states in setting binding legislation, regulatory frameworks, and national targets, as well as the role of NGOs in raising public awareness, advocating for climate justice, and holding both firms and states accountable.

Candidates could also approach the question by assessing the extent to which the profit-driven nature of firms limits their effectiveness in responding to climate change. Candidates could evaluate whether corporate climate actions represent genuine systemic change or superficial strategies (e.g., "greenwashing," voluntary commitments without enforcement). Candidates could contrast corporate initiatives driven by market opportunities (e.g., ESG investments, circular economy models) with instances where short-term profit maximisation leads firms to lobby against environmental regulations or delay decarbonisation. Candidates could also consider how no single actor plays a "key" role in isolation, but rather that a successful climate response relies on the interdependence of firms, states, and NGOs across different contexts and scales. Candidates could analyse how state policies (e.g., carbon pricing, subsidies) create the market incentives necessary for firms to adopt green practices, while NGO pressure mobilises consumer demand and scrutinises corporate practices to enforce transparency.

Levels marked using Generic Level Descriptors for H2 essays

Levels	Marks	Generic Level Descriptors for H2 Essays
5	18–20	Evaluation is consistently analytical and coherent. Response is well-supported by relevant material, including the effective use of examples. Response features accurate geographical knowledge and reflects good understanding of the subject content relevant to the question.
4	14–17	Evaluation is analytical and coherent. Response is mostly well-supported by relevant material, including the appropriate use of examples. Response features accurate geographical knowledge and reflects adequate understanding of the subject content relevant to the question.
3	10–13	Evaluation is broadly analytical and generally coherent. Response is moderately well-supported by relevant material, including some appropriate use of examples. Response features accurate geographical knowledge and reflects adequate understanding of the subject content relevant to the question.
2	6–9	Response is largely descriptive with limited analysis and evaluation. Response is partly coherent and may lack clarity in parts. Response is poorly supported by relevant materials, including the limited use of examples. Response features inaccurate geographical knowledge and poor understanding of the subject content relevant to the question.
1	1–5	Response is descriptive with no analysis or evaluation. Response is fragmented and lacks clarity. Response consists of unsupported assertions. Response features largely inaccurate geographical knowledge and a lack of understanding of the subject content relevant to the question.
0	0	No creditworthy response